

## Practical 8:

**Aim:** A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

## Code:

```
GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[4], time[4];
    int i,j,n=12,frames=4,hit=0,fault=0,counter=0,pos,min;

    for(i=0;i<frames;i++)
        frame[i] = -1;

    for(i=0;i<n;i++) {
        int found = 0;

        for(j=0;j<frames;j++) {
            if(frame[j] == pages[i]) {
                hit++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        if(!found) {
            fault++;

            if(i < frames) {
                frame[i] = pages[i];
                counter++;
                time[i] = counter;
            }
            else {
                min = time[0];
                pos = 0;

                for(j=1;j<frames;j++) {
                    if(time[j] < min) {
                        min = time[j];
                        pos = j;
                    }
                }

                frame[pos] = pages[i];
                counter++;
                time[pos] = counter;
            }
        }
    }

    printf("Hits = %d\nFaults = %d\n", hit, fault);
    printf("Hit %% = %.2f\nFault %% = %.2f\n", (float)hit/n*100, (float)fault/n*100);

    return 0;
}
```

## OUTPUT:-

```
M ~  
  
lenovo@LAPTOP-6627EU0U MSYS ~  
$ nano lu.c  
  
lenovo@LAPTOP-6627EU0U MSYS ~  
$ ls  
bankers.c  bankers.exe  dining.c  dining.exe  lu.c  process  
  
lenovo@LAPTOP-6627EU0U MSYS ~  
$ gcc lu.c -o lu -pthread  
  
lenovo@LAPTOP-6627EU0U MSYS ~  
$ ./lu  
Hits = 4  
Faults = 8  
Hit % = 33.33  
Fault % = 66.67  
  
lenovo@LAPTOP-6627EU0U MSYS ~  
$ |
```